POLITECNICO MILANO 1863 | DIPARTIMENTO DI INGEGNERIA CIVILE E AMBIENTALE

Geographic Information Systems – 2025-26

Installation and usage guide: FTP and FileZilla

Qiongjie Xu

Introduction – Credentials

For the GIS course of 2025/2026, students must develop a laboratory that involves all the practical and theoretical parts of the course, including Desktop GIS, Web GIS, and the technologies and software given during the lessons.

For that regard, we have enabled a GeoServer instance that is reachable at the following link: https://www.gis-geoserver.polimi.it/geoserver/web/?0.

A set of workspaces, usernames and passwords are configured for this GeoServer, one for each group. In case you want to use have access, please send an email to vasil.yordanov@polimi.it and to qiongjie.xu@polimi.it where you specify your group number.

Introduction – FTP

As mentioned in the WebGIS lessons; to upload data to the online GeoServer it is necessary to have the files directly in the server file system. This is possible using the FTP protocol.

In this guide, you will be shown how to connect to the FTP server of the online GeoServer through an FTP client called **FileZilla**, which is free and open-source. Then, a short tutorial will be presented on how to upload and publish your own data in the online GeoServer.

FTP stands for “File Transfer Protocol” and is a communication protocol for the internet that allows to browse and modify the file system of a remote computer. If you want to learn more about FTP and how it works, you can refer to the following link: Tutorial - What is FTP?

Downloading and installing FileZilla

01

1. Access FileZilla website

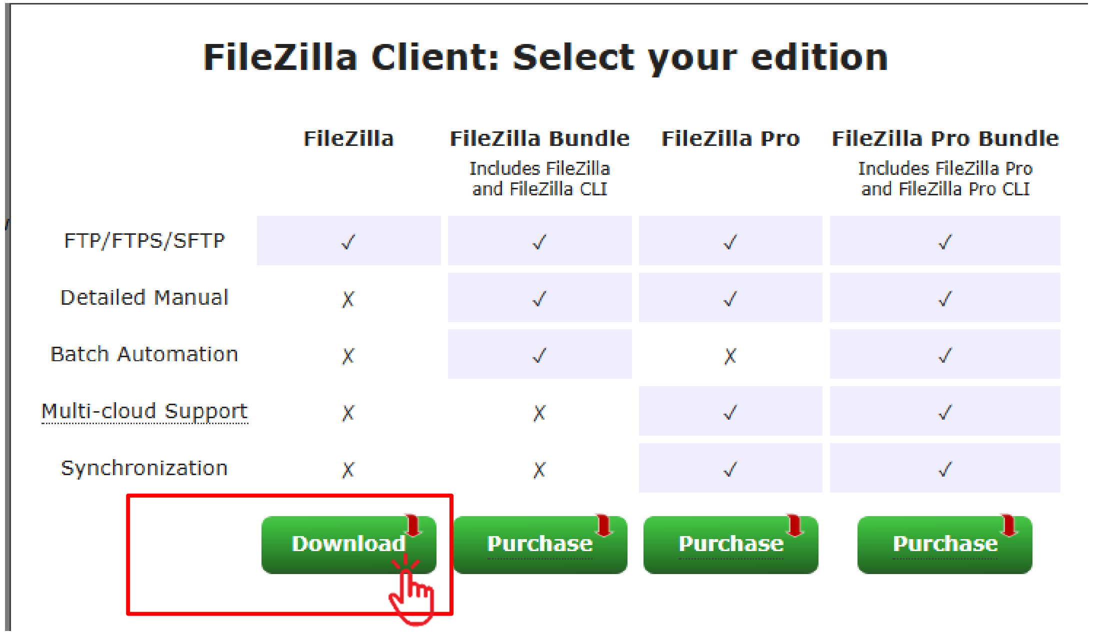
- Go to https://filezilla-project.org/
- From the *Quick downloads links* options, select *Download FileZilla Client*
- As of May 2026, FileZilla Client 3.70.5 is the latest stable version
- FileZilla is a software designed to transfer files between two servers / computers through FTP (*File Transfer Protocol*). FileZilla provides a Graphical User Interface which eases the upload, download, delete, rename and copy files to a remote server.

2. Download FileZilla client for your OS

- For Windows, click on ***Download FileZilla Client***, to download the installation execution file.
    - Windows (64bit x86)

- In case of a different OS, find the corresponding installation file under ***More downloads options*** list.
    - Windows (32bit)
    - MacOS (Intel)
    - MacOS (Apple Silicon - ARM)
    - Linux (64bit)
    - Linux (32bit)

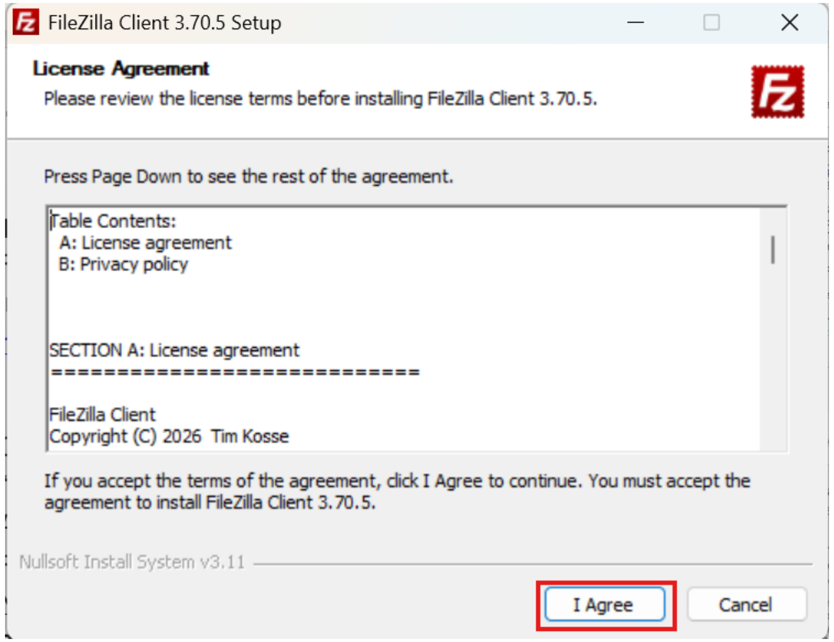
- Download the standard edition for FileZilla Client

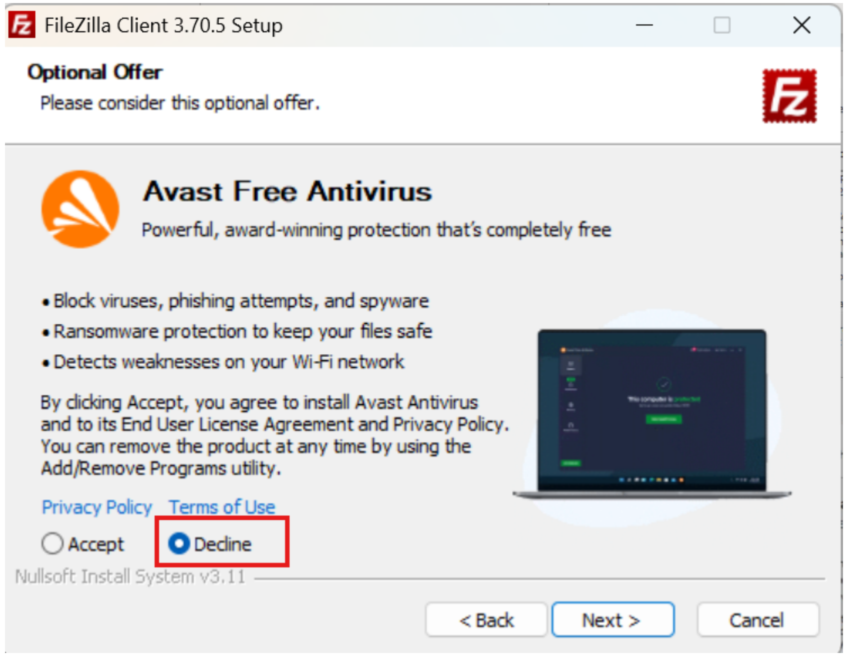


3. Install FileZilla Client

- The installation instructions will only cover Windows.
- For other operating systems, refer to the original installing guide located at: https://wiki.filezilla-project.org/Client_Installation

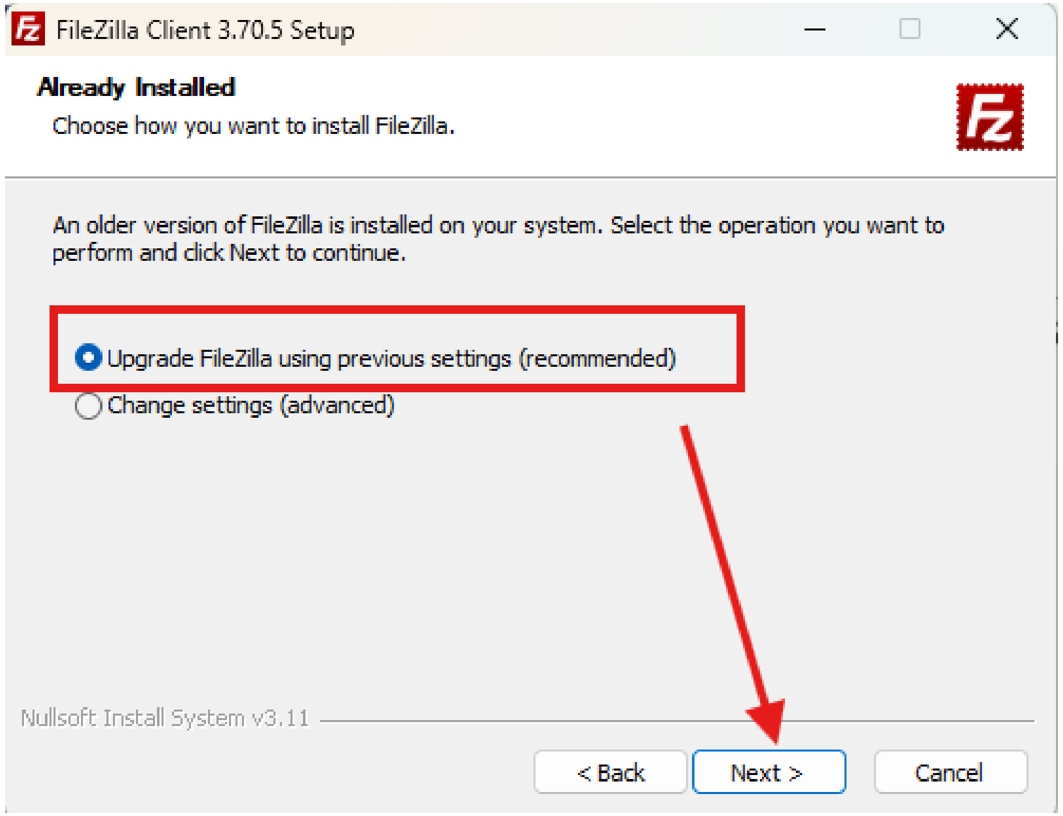
- Open the installation file named "**FileZilla_3.70.5_win64_sponsored2-setup**". Allow the program to make changes to the system.
- Follow the instructions:
- **Note: Make sure to DECLINE the installation any additional installations, such as the Avast Secure Browser**



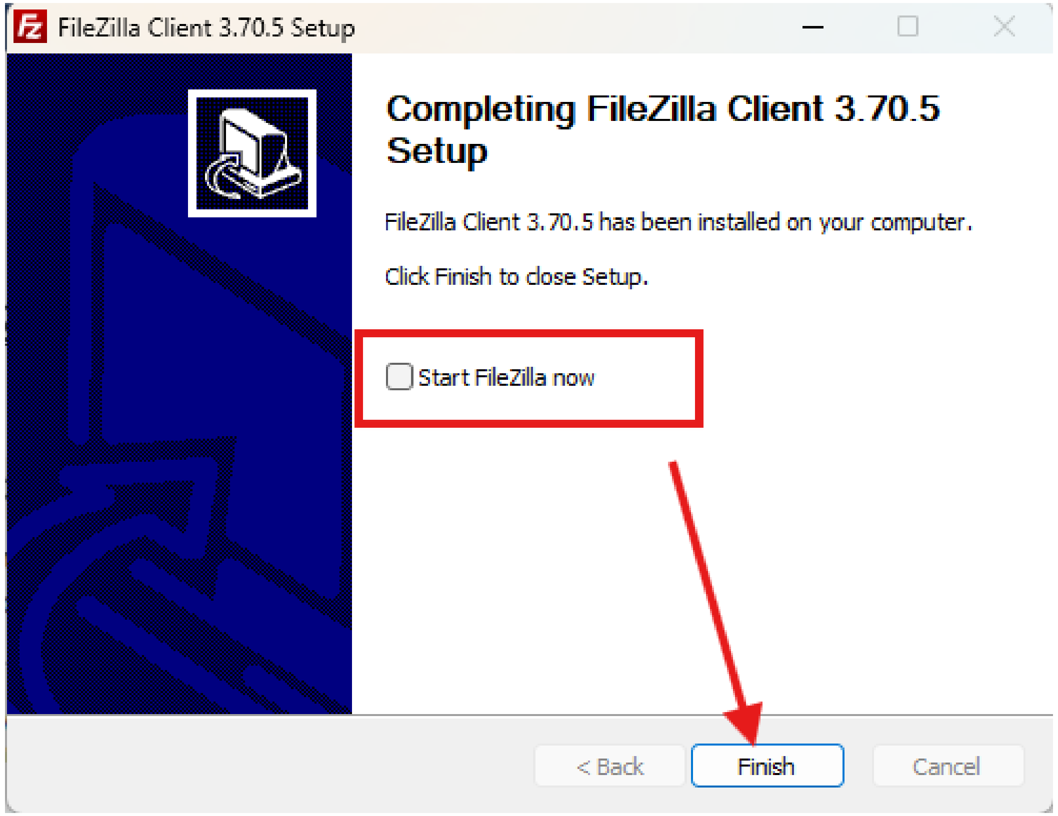


3. Install FileZilla Client

- If FileZilla is already installed, select Upgrade Filezilla using the recommended settings.



- Uncheck the ***Start FileZilla now*** box, and finish to close the installation setup assistant



Using FileZilla

02

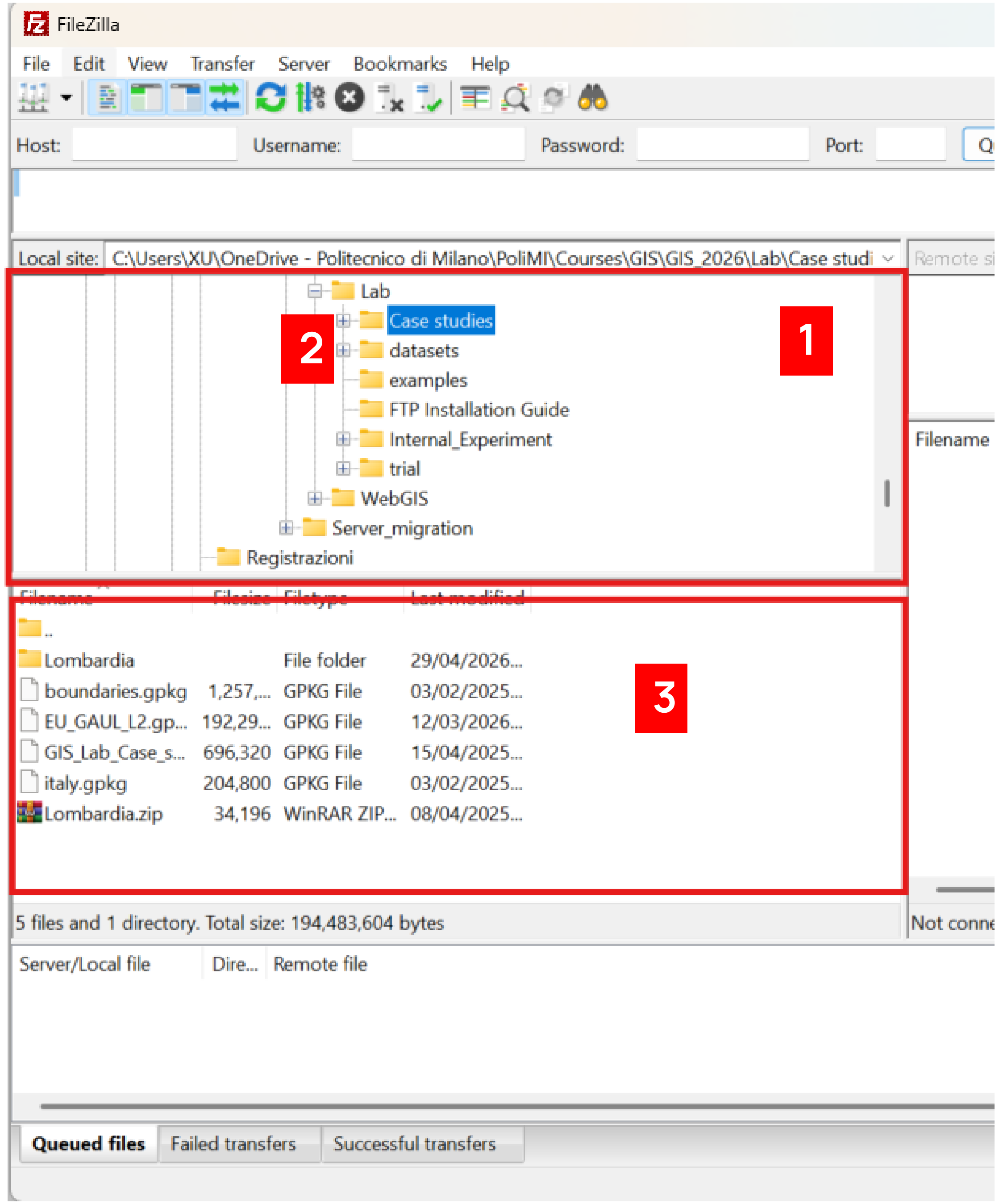
4. FileZilla Client – Local data

- Open the FileZilla Client.

- At first, FileZilla looks like any other file explorer (and it basically is). You can browse over your own files. Search the data that you want to upload to the online GeoServer with the following steps:

- On the Graphical User Interface (GUI), from the *Local directory tree* **(1)**, find the path to your local data directory and select it **(2)**

- Once the data directory is selected, the contents of the directory will appear in the *Local transfer window* below the *Local directory tree* **(3)**



5. FileZilla Client – Connecting to the FTP Server

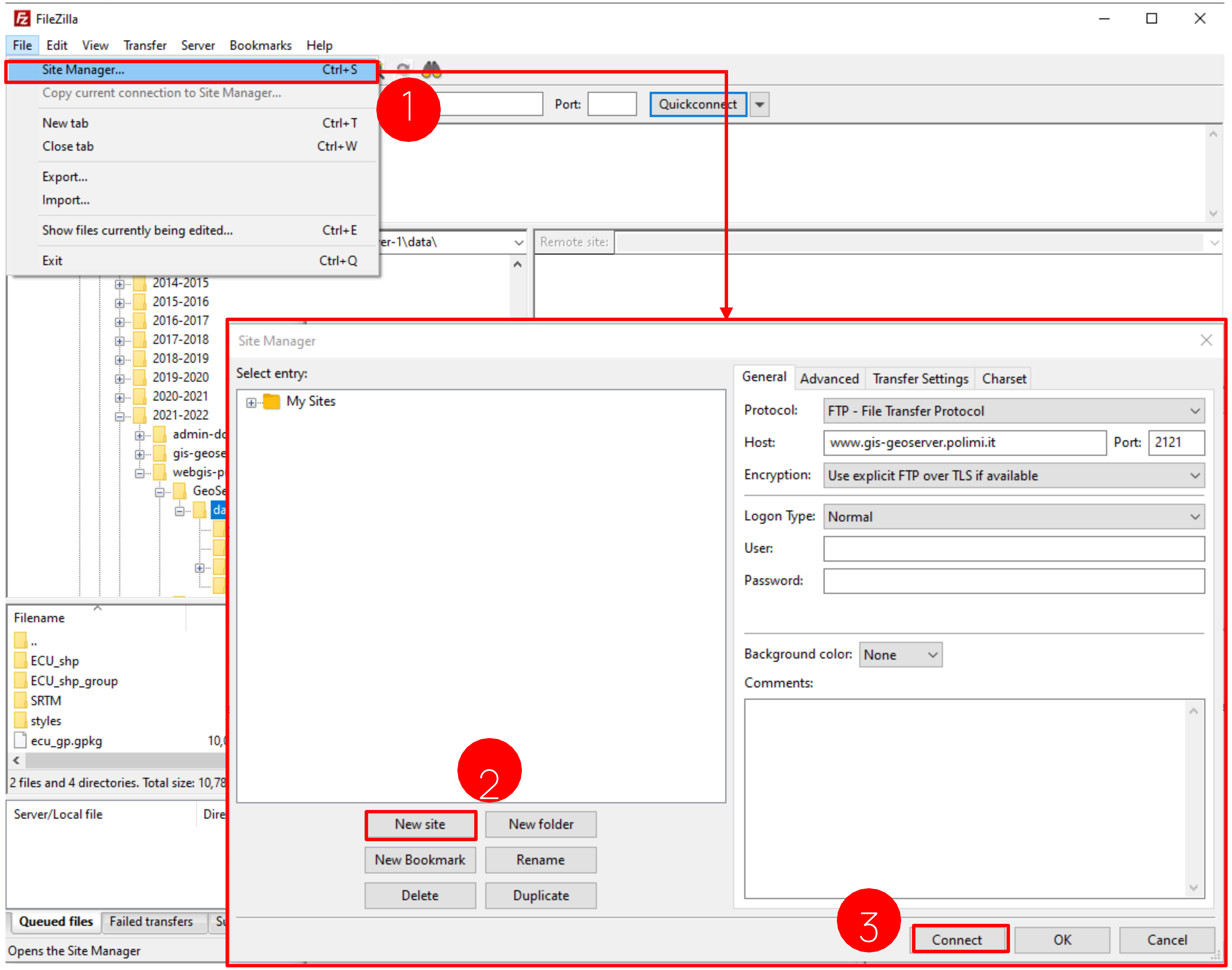
- *We must connect to the online GeoServer FTP server to put our files into it.*

  File ➔ *Site Manager…* **(1)** ➔ on the prompted window, select *New site* **(2)** and proceed to compile the form

  - Protocol: FTP – File Transfer Protocol
  - Host: www.gis-geoserver.polimi.it
  - Port: 2121
  - Encryption: Use explicit FTP over TLS if available
  - Logon Type: Normal
  - User: {group user}
  - Password: {group password}

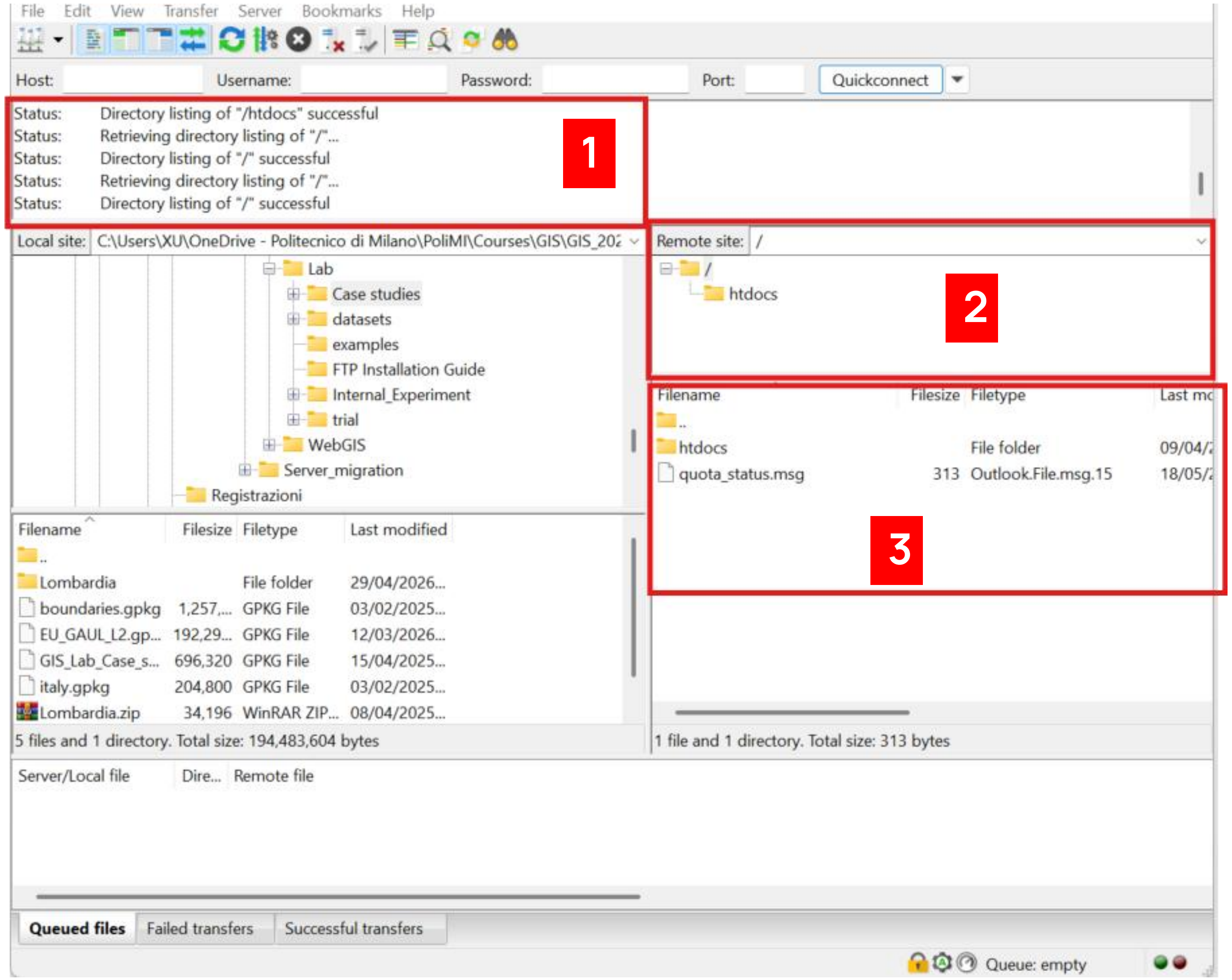
- *Connect* **(3)** ➔ Verify that you trust the server certificate by clicking *OK*

- **Please, keep your FTP credentials privately!** Each group has been assigned a personal directory for the management of the files for the project. Only group members must be allowed to modify the contents of the directory.



6. FileZilla Client – Upload Files

- If the connection to the remote server works, the ***Message log* (1)** should indicate the following success message:
  - Status: "Directory listing of "/" successful"
- Also, the ***Remote directory tree*** should present the contents of the connected directory **(2).** Under the ***Remote directory tree,*** there is the ***Remote transfer window* (3)** displaying the contents of the currently selected folder of the ***Remote directory tree.***
- The remote directory that is visible to the GeoServer instance (https://www.gis-geoserver.polimi.it/geoserver) is inside the ***htdocs*** folder



6. FileZilla Client – Upload Files

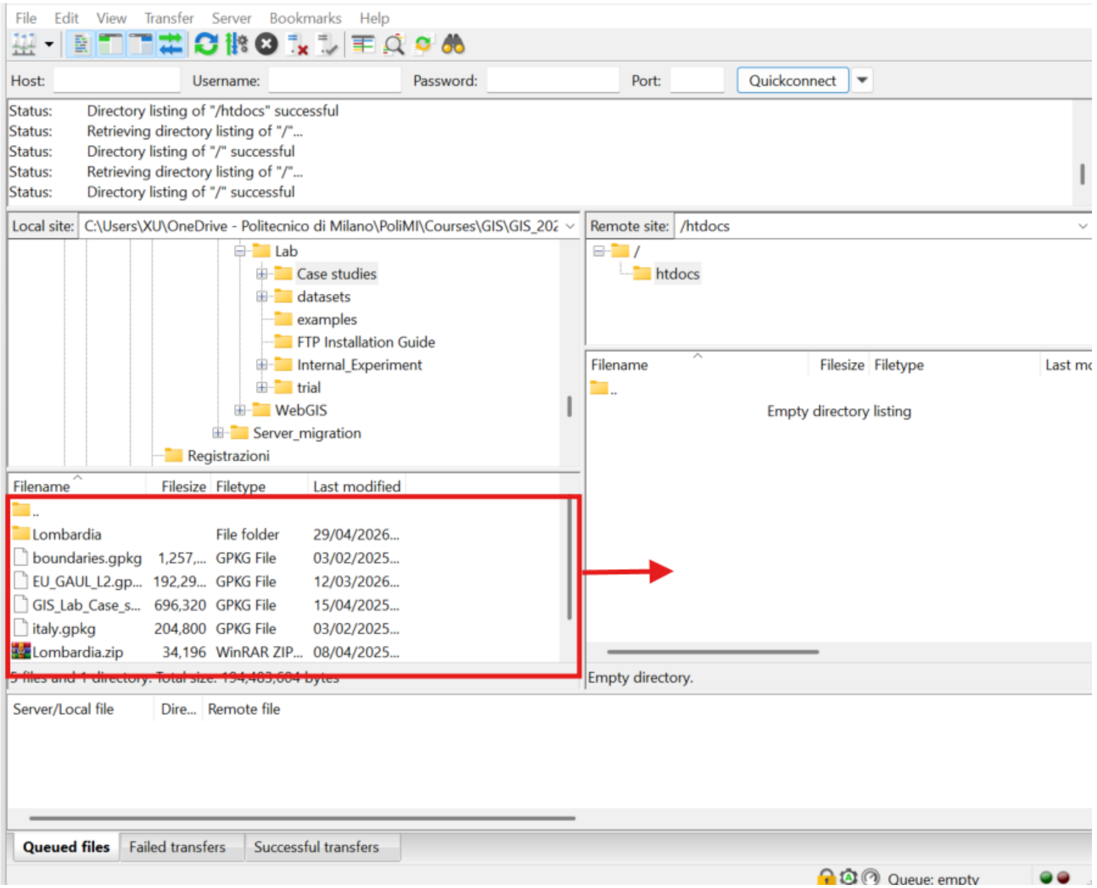
- Inside the ***htdocs*** folder, it will be possible to upload the datasets for the project.

- To add files from your local computer, drag-and-drop the files from the ***Local transfer Window*** To the ***Remote transfer window.***

- The datasets will be available from the GeoServer Web Administrative Interface, under the location:

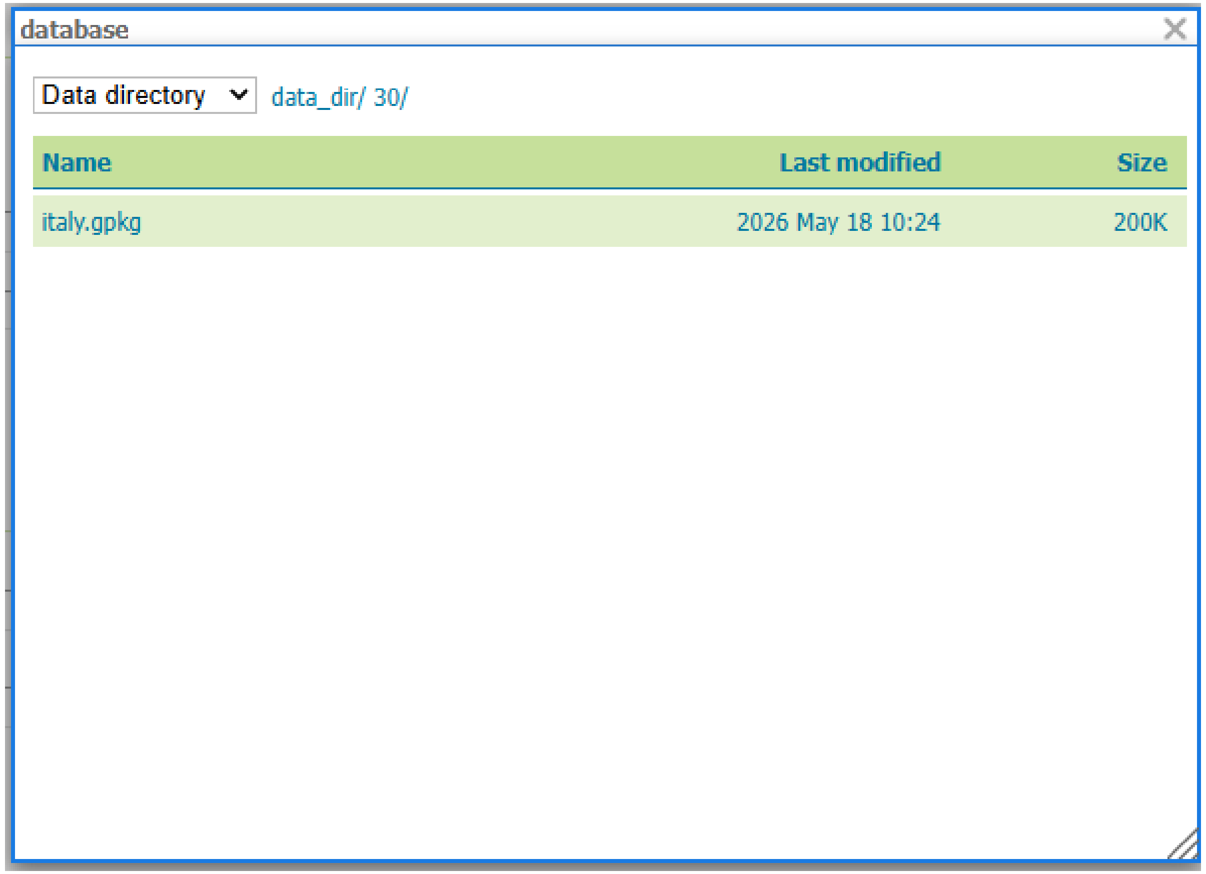
    - /data_dir/{group_number}

- From this location, it is possible to use the layers inside GeoServer.



7. Locating the files in GeoServer

- Access the online GeoServer using the **same username and password** that you used to connect in FileZilla.
- Now, the files that you uploaded through FileZilla should be available inside **"data_dir/{group number}"**
    - You can create a new store and navigate the data directory to locate your files!
- **NOTE: Each group has 1GB of space available inside the online GeoServer. Use it wisely!**



Support

For any technical issues or questions, please, send an email to:

qiongjie.xu@polimi.it